Differential Manifolds

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Chapter I

Manifolds

§1 Manifolds

§1.1 Topological Manifolds

Definition 1.1. *Suppose M is a topological space. We say that M is a* ***topological manifold*** *of dimension n or a topological n-manifold if it has the following properties:*

(i) *M is a Hausdorff space*

(ii) *M is second-countable*

(iii) *M is locally Euclidean of dimension n: each point p of M has a neighborhood U that is homeomorphic to an open subset $\widehat{U}$ of $\mathbb{R}^n$. (U, φ) called* ***coordinate chart****.*

Given a chart (U, φ), we call the set U a ***coordinate domain****, and φ is called a* ***local coordinate map****.*

§1.2 Topological Properties of Manifolds

Proposition 1.2. *Suppose M is a topological manifold, then*

(1) M is local compact

(2) M is connected

(4) M is path-connected

(5) M is Lindelof's : any open

Definition 1.3. *Let M be a topological space. A collection $\mathcal{X}$ of subsets of M is said to be* ***locally finite*** *if each point of M has a neighborhood that intersects at most finitely many of the sets in $\mathcal{X}$.*

Given a cover $\mathcal{U}$ of M, another cover $\mathcal{V}$ is called a ***refinement*** *of $\mathcal{U}$ if for each $V \in \mathcal{V}$ there exists some $U \in \mathcal{U}$ such that $V \subseteq U$.*

We say that M is ***paracompact*** *if every open cover of M admits an open, locally finite refinement.*

Theorem 1.4 (σ-compact)**.** *A second-countable, locally compact Hausdorff space (thus a manifold) admits an exhaustion by compact sets.*

Proof: Let X be such a space. Because X is a locally compact Hausdorff space, it has a basis of precompact open subsets; since it is second-countable, it is covered by countably many such sets. Let $(U_i)_{i=1}^{\infty}$ be such a countable cover. Beginning with $K_1 = \overline{U}_1$, assume by induction that we have constructed compact sets $K_1, \ldots, K_k$ satisfying $U_j \subseteq K_j$ for each j and $K_{j-1} \subseteq \operatorname{Int} K_j$ for $j \geq 2$. Because K_k is compact, there is some m_k such that $K_k \subseteq U_1 \cup \cdots \cup U_{m_k}$. If we let $K_{k+1} = \overline{U}_1 \cup \cdots \cup \overline{U}_{m_k}$, then K_{k+1} is a compact set whose interior contains K_k. Moreover, by increasing m_k if necessary, we may assume that $m_k \geq k+1$, so that $U_{k+1} \subseteq K_{k+1}$. By induction, we obtain the required exhaustion.

Theorem 1.5 (Paracompact)**.** *Every topological manifold is paracompact. In fact, given a topological manifold M, an open cover $\mathcal{X}$ of M, and any basis $\mathfrak{B}$ for the topology of M, there exists a countable, locally finite open refinement of $\mathcal{X}$ consisting of elements of $\mathcal{B}$.*

Proof: Given $M, \mathcal{X}$, and $\mathcal{B}$ as in the hypothesis of the theorem, let $(K_j)_{j=1}^{\infty}$ be an exhaustion of M by compact sets (Proposition A.60). For each j, let $V_j = K_{j+1} \setminus \operatorname{Int} K_j$ and $W_j = \operatorname{Int} K_{j+2} \setminus K_{j-1}$ (where we interpret K_j as $\varnothing$ if $j < 1$). Then V_j is a compact set contained in the open subset W_j. For each $x \in V_j$, there is some $X_x \in \mathcal{X}$ containing x, and because $\mathcal{B}$ is a basis, there exists $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq X_x \cap W_j$. The collection of all such sets B_x as x ranges over V_j is an open cover of V_j, and thus has a finite subcover. The union of all such finite subcovers as j ranges over the positive integers is a countable open cover of M that refines $\mathcal{X}$. Because the finite subcover of V_j consists of sets contained in W_j, and $W_j \cap W_{j'} = \varnothing$ except when $j-2 \leq j' \leq j+2$, the resulting cover is locally finite.

§1.3 Smooth Structures and Smooth Manifolds

Definition 1.6. *Let M be a topological n-manifold.*

1. *Two charts (U, φ) and (V, ψ) are said to be **smoothly compatible** if either $U \cap V = \varnothing$ or the two **transition map***

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V), \quad \varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$$

   are C^{∞}.

2. *We define an **smooth atlas** $\mathcal{A}$ for M to be a collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ whose domains cover M and any two charts in $\mathcal{A}$ are smoothly compatible with each other.*

3. *A smooth atlas $\mathcal{A}$ on M is **maximal** if it is not properly contained in any larger smooth atlas. If M is a topological manifold, a **smooth structure** on M is a maximal smooth atlas.*

   *A C^{∞} manifold is a pair $(M, \mathcal{A})$, where M is a topological manifold and $\mathcal{A}$ is a **smooth structure** on M.*

Although the C^∞ compatibility of charts is clearly reflexive and symmetric, it is not transitive. However, we have the following lemma:

Lemma 1.7. *Let $\{(U_\alpha, \phi_\alpha)\}$ be an atlas on a locally Euclidean space. If two charts (V, ψ) and (W, σ) are both compatible with the atlas $\{(U_\alpha, \phi_\alpha)\}$, then they are compatible with each other.*

Proposition 1.8. *Let M be a topological manifold.*

1. *Every smooth atlas $\mathcal{A}$ for M is contained in a unique maximal smooth atlas $\overline{\mathcal{A}}$, called the* ***smooth structure determined by*** *$\mathcal{A}$. Indeed, $\overline{\mathcal{A}}$ denote the set of all charts that are smoothly compatible with every chart in $\mathcal{A}$.*

2. *Two smooth atlases for M determine the same smooth structure if and only if their union is a smooth atlas.*

Definition 1.9. *A subset S of a C^∞ manifold M of dimension n is a* ***regular submanifold of dimension*** *s if for every $p \in S$ there is a coordinate neighborhood (U, φ) of p such that $U \cap S$ is defined by the vanishing of $n - s$ of the coordinate functions. By renumbering the coordinates, we may assume that these $n - k$ coordinate functions are $x^{s+1}, \ldots, x^n$.*

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^s \times 0)$$

Let

$$\varphi_S : U \cap S \to \mathbb{R}^s$$

be the restriction of the first k components of φ to $U \cap S$. Then $(U \cap S, \varphi_S)$ is a chart for S in the subspace topology.

§1.4 Manifold with Boundary

§1.5 Topological Manifold with Boundary

Definition 1.10. *An* ***n-dimensional topological manifold with boundary*** *is a second-countable Hausdorff space M in which every point p has a neighborhood U homeomorphic φ to an open subset of $\mathbb{R}^n_+$. The open subset $U \subseteq M$ together with a map $\varphi : U \to \mathbb{R}^n_+$ that is a homeomorphism onto an open subset of $\mathbb{R}^n_+$ will be called a* ***chart*** *for M.*

We will call (U, φ) an ***interior chart*** *if $\varphi(U)$ is an open subset of $\mathbb{R}^n$, and a* ***boundary chart*** *if $\varphi(U)$ is an open subset of $\mathbb{R}^n_+$ such that $\varphi(U) \cap \partial\mathbb{R}^n_+ \neq \varnothing$.*

A point $p \in M$ is called an ***interior point*** *of M if it is in the domain of some interior chart. It is a* ***boundary point*** *of M if it is in the domain of a boundary chart that sends p to $\partial\mathbb{R}^n_+$.*

The boundary of M (the set of all its boundary points) is denoted by ∂M; similarly, its interior, the set of all its interior points, is denoted by $\operatorname{Int} M$.

Remark *∂M and* $\operatorname{Int} M$ *are disjoint sets whose union is M*

Proposition 1.11. *Let M be a topological n-manifold with boundary.*

(1) $\operatorname{Int} M$ *is an open subset of M and a topological n-manifold without boundary.*

(2) ∂M is a closed subset of M and a topological $(n-1)$-manifold without boundary.

(3) M is a topological manifold if and only if $\partial M = \varnothing$.

(4) If $n = 0$, then $\partial M = \varnothing$ and M is a 0 -manifold.

Proposition 1.12. *Let M be a topological manifold with boundary.*

(1) M has a countable basis of precompact coordinate balls and half-balls.

(2) M is locally compact.

(3) M is paracompact.

(4) M is locally path-connected.

(5) M has countably many components, each of which is an open subset of M and a connected topological manifold with boundary.

(6) The fundamental group of M is countable.

§1.6 Smooth Manifold with Boundary

Definition 1.13. *Let M be a topological manifold with boundary. A **smooth structure** for M is defined to be a maximal smooth atlas $\mathcal{A}$, a collection of charts whose domains cover M and whose transition maps (and their inverses) are smooth. With such a structure, $(M, \mathcal{A})$ is called a **smooth manifold with boundary**.*

Just as for smooth manifolds, if M is a smooth manifold with boundary, any chart in the given smooth atlas is called a smooth chart for M.

Recall that a map from an arbitrary subset $A \subseteq \mathbb{R}^n$ to $\mathbb{R}^k$ is said to be smooth if in a neighborhood of each point of A it admits an extension to a smooth map defined on an open subset of $\mathbb{R}^n$.

§2 Smooth Maps on a Manifold

§2.1 Smooth Functions on a Manifold

Definition 2.1. *Let M be a smooth n-manifold, k is a nonnegative integer, and $f : M \to \mathbb{R}^k$ is any function. We say that f is **smooth at** p if there exists a chart (U, φ) for M whose domain contains p and such that the composite function*

$$f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^k$$

*is smooth at $\varphi(p)$ (i.e. smooth on a neighborhood of $\varphi(p)$ in $\varphi(U)$). The function $f \circ \varphi^{-1}$ is called the **local coordinate representation** of f. The function f is said to be C^∞ on M if it is C^∞ at every point of M.*

Remark. *The definition of the smoothness of a function f at a point is independent of the chart (U, φ).*

Proposition 2.2. *Let M be a manifold of dimension n, and $f : M \to \mathbb{R}$ a real-valued function on M. The following are equivalent:*

1. *The function $f : M \to \mathbb{R}$ is C^∞.*
2. *The manifold M has an atlas such that for every chart (U, ϕ) in the atlas, $f \circ \phi^{-1} : \mathbb{R}^n \supset \phi(U) \to \mathbb{R}$ is C^∞.*
3. *For every chart (V, ψ) on M, the function $f \circ \psi^{-1} : \mathbb{R}^n \supset \psi(V) \to \mathbb{R}$ is C^∞.*

Proposition 2.3 (Smoothness in terms of components)**.** *Let M be a manifold. A vector-valued function $F : M \to \mathbb{R}^m$ is C^∞ if and only if its component functions $F^1, \ldots, F^m : M \to \mathbb{R}$ are all C^∞.*

§2.2 Smooth Maps Between Manifolds

Definition 2.4. *Let M, N be smooth manifolds, and let $f : M \to N$ be a continuous map. We say that f is a* ***smooth map*** *if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $f(p)$ (assume that $f(U) \subseteq V$ without generality) and the composite map*

$$\tilde{f} = \psi \circ f \circ \varphi^{-1} : \varphi(f^{-1}(V) \cap U) \to \psi(V)$$

is smooth. We call $\tilde{f} = \psi \circ f \circ \varphi^{-1}$ the ***coordinate representation of f with respect to the given coordinates****.*

Proposition 2.5. *Let N and M be smooth manifolds, and $F : N \to M$ a continuous map. The following are equivalent:*

1. *The map $F : N \to M$ is C^∞.*
2. *There are atlases $\mathfrak{U}$ for N and $\mathfrak{V}$ for M such that for every chart (U, ϕ) in $\mathfrak{U}$ and (V, ψ) in $\mathfrak{V}$, the map*
$$\psi \circ F \circ \phi^{-1} : \phi\left(U \cap F^{-1}(V)\right) \to \mathbb{R}^m$$
is C^∞.
3. *For every chart (U, ϕ) on N and (V, ψ) on M, the map*
$$\psi \circ F \circ \phi^{-1} : \phi\left(U \cap F^{-1}(V)\right) \to \mathbb{R}^m$$
is C^∞.

Proposition 2.6 (Composition of C^∞ maps)**.** *Let M, N, and P be smooth manifolds with or without boundary.*

1. *Every constant map $M \to N$ is smooth.*
2. *The identity map of M is smooth.*
3. *If $U \subseteq M$ is an open submanifold with or without boundary, then the inclusion map $U \hookrightarrow M$ is smooth.*
4. *If $F : M \to N$ and $G : N \to P$ are smooth, then so is $G \circ F : M \to P$.*

Definition 2.7. *If M and N are smooth manifolds with or without boundary, a **diffeomorphism** from M to N is a smooth bijective map $F : M \to N$ that has a smooth inverse. We say that M and N are **diffeomorphic** if there exists a diffeomorphism between them. Sometimes this is symbolized by $M \approx N$.*

Proposition 2.8. *Let U be an open subset of a manifold M of dimension n and $f : U \to f(U) \subset \mathbb{R}^n$ be a maps. Then f is a diffeomorphism onto its image if and only if (U, f) is a coordinate chart in the differentiable structure of M.*

§2.3 Partial Derivatives and the Rank of a Smooth Map

On a manifold M of dimension m, let (U, ϕ) be a chart. As a function into $\mathbb{R}^m$, ϕ has m components $x^1, \ldots, x^m$ defined by

$$\phi(p) = \left(x^1(p), \ldots, x^m(p)\right)$$

This means that if $r^1, \ldots, r^m$ are the standard coordinates on $\mathbb{R}^m$, then $x^i = r^i \circ \phi$.

Definition 2.9. *Let M be a manifold of dimension m, and let $(U, \varphi) = (U, x^1, \ldots, x^m)$ be a chart on M.For $f \in C^\infty(U, \mathbb{R})$ and a point $p \in M$, we define the **partial derivative of** f **with respect to** x^i **at** p to be*

$$\frac{\partial(f \circ \varphi^{-1})}{\partial r^i}(\varphi(p))$$

Remark. *Usually written as $\frac{\partial}{\partial x^i}\Big|_p f$ or $\frac{\partial f}{\partial x^i}(p)$. Thus*

$$\frac{\partial}{\partial x^i}\bigg|_p f = \frac{\partial f}{\partial x^i}(p) = \frac{\partial\left(f \circ \phi^{-1}\right)}{\partial r^i}(\phi(p)) = \frac{\partial}{\partial r^i}\bigg|_{\phi(p)} \left(f \circ \phi^{-1}\right).$$

on U, and

$$\frac{\partial f}{\partial x^i} \circ \phi^{-1} = \frac{\partial\left(f \circ \phi^{-1}\right)}{\partial r^i}.$$

on $\phi(U)$.

Definition 2.10. *Let $f : N \to M$ be a smooth map, and let $(U, \varphi) = (U, x^1, \ldots, x^n)$ and $(V, \psi) = (V, y^1, \ldots, y^m)$ be charts on M and N respectively such that $f(U) \subset V$. Denote by*

$$f^i := y^i \circ f : U \to \mathbb{R}$$

Then the matrix

$$\left[\frac{\partial f^i}{\partial x^j}\right]_{\varphi(p)}$$

is called the ***Jacobian matrix of*** f ***relative to the charts*** (U, φ) ***and*** (V, ψ). *It is also written as* $\partial\left(f^1, \ldots, f^n\right) / \partial\left(x^1, \ldots, x^m\right)$. *We define the rank of* f *at* p

$$\operatorname{rank}_p f := \operatorname{rank}\left(\frac{\partial f^i}{\partial x^j}\right)_{\varphi(p)}$$

Remark. *The rank of* f *at* p *is independent of the choice of charts.*

Theorem 2.11 (Constant rank theorem)**.** *Let* M *and* N *be manifolds of dimensions* m *and* n *respectively. Suppose* $f : M \to N$ *has constant rank* k *in a neighborhood of a point* p *in* M. *Then there are charts* (U, φ) *centered at* p *in* N *and* (V, ψ) *centered at* $f(p)$ *in* M *such that for* $(x^1, \ldots, x^m)$ *in* $\varphi(U)$,

$$\psi \circ f \circ \varphi^{-1} : \left(x^1, \ldots, x^m\right) \mapsto \left(x^1, \ldots, x^k, 0, \ldots, 0\right)$$

§3 Quotients

§4 Partition of unity

§4.1 Topological Preliminaries

Definition 4.1. *Let* X *be a topological space.*

(1) A collection $\mathcal{A}$ *of subsets of* X *is said to be* ***locally finite*** *in* X *if every point of* X *has a neighborhood that intersects only finitely many elements of* $\mathcal{A}$.

(2) A collection $\mathcal{B}$ *of subsets of* X *is said to be* ***countably locally finite*** *if* $\mathcal{B}$ *can be written as the countable union of collections* $\mathcal{B}_n$, *each of which is locally finite.*

Definition 4.2. *Let* X *be a topological space, if there exists* X_j *such that*

(i) For each j, *the closure* $\overline{X}_j$ *is compact.*

(ii) For each j, $\overline{X}_j \subset X_{j+1}$.

(iii) $M = \bigcup_j X_j$.

The subsets $\{X_j\}$ *described is called an* ***exhaustion*** *of* M.

Definition 4.3. *A real-valued continuous function* f *on* X *is called an* ***exhausion function*** *for* X *if for any* $c \in \mathbb{R}$, *the sublevel set* $f^{-1}((\infty, c])$ *is compact.*

Lemma 4.4 (Lindelof's)**.** *Let* X *be a second countable space, then every open cover* $\mathcal{A}$ *has a countable subcover.*

Theorem 4.5. *If X is a second countable, locally compact and Hausdorff space (thus a manifold), then there exists a exhaustion of X.*

Proof. First, there exists a open cover $\mathcal{A}$ that every element of $\mathcal{A}$ has compact closure, then there exists countable many

$$A_1, A_2, \ldots$$

such that $\bigcup A_i = X$ and $\overline{A_i}$ is compact.

We let $X_1 = A_1$. Since A_i is an open cover of $\overline{X}_1$ which is compact, there exists finitely many open sets $A_{i_1}, \cdots, A_{i_k}$ so that $\overline{X}_1 \subset A_{i_1} \cup \cdots \cup A_{i_k}$. Let $X_2 = A_{i_1} \cup \cdots \cup A_{i_k} \cup A_2$. Obviously $\overline{X}_2$ is compact. Repeat this procedure again and again, we could get a desired sequence of open sets $X_1, X_2, X_3, \cdots$.

Definition 4.6. *Let $\mathcal{A}$ be a open cover of the space X, then a open cover $\mathcal{B}$ of X is said to be a* ***refinement*** *of $\mathcal{A}$ (or is said to refine $\mathcal{A}$) if for each element B of $\mathcal{B}$, there is an element A of $\mathcal{A}$ containing B.*

Definition 4.7. *A space X is* ***paracompact*** *if every open cover $\mathcal{A}$ of X has a locally finite open refinement $\mathcal{B}$ of $\mathcal{A}$.*

Theorem 4.8. *Let M be any topological manifold. For any open cover $\mathcal{U} = \{U_\alpha\}$ of M, one can find two countable family of open covers $\mathcal{V} = \{V_j\}$ and $\mathcal{W} = \{W_j\}$ of M so that*

(i) $\mathcal{W}$ is a locally finite open refinement of $\mathcal{U}$.

(ii) For each j, $\overline{V}_j$ is compact and $\overline{V}_j \subset W_j$.

Proof. Let $\{X_j\}$ be a exhaustion of M. For each $p \in M$, there is an j and an $\alpha(p)$ so that $p \in \overline{X}_{j+1} \backslash X_j$ and $p \in U_{\alpha(p)}$. Since M is locally Euclidean, one can always choose open neighborhoods V_p, W_p of p so that $\overline{V}_p$ is compact and

$$p \in V_p \subset \overline{V}_p \subset W_p \subset U_{\alpha(p)} \cap \left(X_{j+2} \backslash \overline{X}_{j-1}\right)$$

Now for each j, since the "stripe" $\overline{X}_{j+1} \backslash X_j$ is compact, one can choose finitely many points $p_1^j, \cdots, p_{k_j}^j$ so that $V_{p_1^j}, \cdots, V_{p_{k_j}^j}$ is an open cover of $\overline{X}_{j+1} \backslash X_j$. Denote all these $V_{p_k^j}$'s by $V_1, V_2, \cdots$, and the corresponding $W_{p_k^j}$'s by $W_1, W_2, \cdots$. Then $\mathcal{V} = \{V_k\}$ and $\mathcal{W} = \{W_k\}$ are open covers of M that satisfies all the conditions.

Corollary 4.9. *Manifolds is paracompact, σ-compact*

§4.2 Existence

Lemma 4.10. *There exists f_1, f_2, f_3 such that*

Theorem 4.11 (Bump function on manifold)**.** *Let M be a smooth manifold, $K \subset M$ is a compact subset, and $U \subset M$ an open subset that contains K. Then there is a $\varphi \in C^\infty(M)$ so that $K \prec f \prec U$ ($0 \leq \varphi \leq 1, \varphi \equiv 1$ on K and* $\operatorname{supp}(\varphi) \subset U$*).*

Proof. For each $q \in K$, there is a chart (φ_q, U_q, V_q) near q so that $U_q \subset U$ and V_q contains the open ball $B_3(0)$ in $\mathbb{R}^n$. Let $U'_q = \varphi_q^{-1}(B_1(0))$, and let

$$f_q(p) = \begin{cases} f_3(\varphi_q(p)) & , p \in U_q \\ 0 & , p \notin U_q \end{cases}$$

Then $f_q \in C^\infty(M)$, $\operatorname{supp}(f_q) \subset U_q \subset U$ and $f_1 \equiv 1$ on U'_q.

Now the family of open sets $\{U'_q\}$ is an open cover of K. Since K is compact, there is a finite sub-cover $\{U'_{q_1}, \cdots, U'_{q_n}\}$. Let

$$\psi = \sum_{i=1}^{n} f_{q_i}$$

Then ψ is a smooth and compactly supported function on M so that $\psi \geq 1$ on K and $\operatorname{supp}(\psi) \subset U$. It follows that the function $\varphi(p) = f_2(\psi(p))$ satisfies all the conditions we required.

Definition 4.12. *Suppose M is a topological space, and let $\mathcal{A} = \{A_\alpha\}$ be an arbitrary open cover of M. A **partition of unity subordinate to** $\mathcal{A}$ is an indexed family*

$$\{\rho_\alpha : \rho_\alpha : M \to \mathbb{R} \text{ is continue}\}$$

with the following properties:

(i) $\rho_\alpha \prec A_\alpha$ $(\operatorname{supp} \rho_\alpha \subset A_\alpha)$ *for all* α

(ii) The family of supports $\{\operatorname{supp} \rho_\alpha\}$ *is locally finite.*

(iii) $\sum_\alpha \rho_\alpha(x) \equiv 1$ *on* M

Theorem 4.13. *Suppose M is a smooth manifold with or without boundary, and $\{U_\alpha\}_{\alpha \in A}$ is any indexed open cover of M. Then there exists a smooth partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$.*

Proof. We can find nonnegative functions $\varphi_j \in C^\infty(M)$ so that $\overline{V}_j \prec \varphi_j \prec W_j$ on $\overline{V}_j$ and $\operatorname{supp}(\varphi_j) \subset W_j$. Since $\mathcal{W}$ is a locally finite cover, $\varphi = \sum \varphi_j$ is a well-defined smooth function on M. Since each φ_j is nonnegative, and $\mathcal{V}$ is a cover of M, φ is strictly positive on M. It follows that the functions $\psi_j = \frac{\varphi_j}{\varphi}$ are smooth and satisfy $0 \leq \psi_j \leq 1$ and $\sum_j \psi_j = 1$.

Let

$$\rho_\alpha = \sum_{W_j \subset U_\alpha} \psi_j$$

Note that the right hand side is a finite sum near each point, so it does define a smooth function. Clearly the family $\{\rho_\alpha\}$ is a partition of unity subordinate to $\{U_\alpha\}$.

§4.3 Application

Theorem 4.14 (Smooth Urysohn lemma)**.** *Let M be a smooth manifold, $F \subset M$ is a closed subset, and $U \subset M$ an open subset that contains F. Then there is a "bump" function $\varphi \in C^\infty(M)$ so that $0 \leq \varphi \leq 1, \varphi \equiv 1$ on F and $\operatorname{supp}(\varphi) \subset U$.*

Proof. *Let $\{\rho_1, \rho_2\}$ be a partition of unity subordinate to the open cover $\{U, M\backslash F\}$. Then $\varphi = \rho_1$ is what we need: $\rho_1 = 1$ on F since $\rho_2 = 0$ on F.*

Theorem 4.15 (Whitney Approximation Theorem)**.** *Let M be a smooth manifold, $F \subset M$ a closed subset and k be a positive integer. Then for any continuous function $f : M \to \mathbb{R}^k$ which is smooth on F and any positive continuous function $\delta : M \to \mathbb{R}_{>0}$, there exists $f \in C^\infty(M)$ so that*

$$f(p) = g(p), \quad \forall p \in F$$

and

$$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

Proof. *By definition, there exists an open set $U \supset A$ and a smooth function f_0 defined on U so that $f_0 = f$ on F. Let*

$$U_0 = \{p \in U : |f_0(p) - f(p)| < \delta(p)\}$$

Then U_0 is open in M and $U_0 \supset F$.

Next we construct a open cover of $M\backslash F$. For any $q \in M\backslash F$, we let

$$U_q = \{p \in M\backslash A : |f(p) - f(q)| < \delta(p)\}$$

Then $\{U_q \mid q \in M\backslash F\}$ is an open covering of $M\backslash F$. Now let $\{\rho_0, \rho_q : q \in M\}$ be P.O.U. subordinate to the open cover $\{U_0, U_q : q \in M\}$ of M, and define a function on M via

$$g(p) = \rho_0(p) f_0(p) + \sum_{q \in M} \rho_q(p) f(q).$$

Since the summation is locally finite, g is smooth. Also by definition, $g = f_0 = f$ on F. Moreover, for any $q \in M$ one has

$$\begin{aligned}
|g(p) - f(p)| &= \left| \rho_0(p) f_0(p) + \sum_q \rho_q(p) f(q) - \rho_0(p) f(p) - \sum_q \rho_q(p) f(p) \right| \\
&\leq \rho_0(p) \left| f_0(p) - f(p) \right| + \sum_q \rho_q(p) |f(q) - f(p)| \\
&< \rho_0(p)\delta(p) + \sum_q \rho_q(p)\delta(p) \\
&= \delta(p)
\end{aligned}$$

Corollary 4.16 (Tietze)**.** *Let M be a smooth manifold and closed set $F \subset$. If f is smooth on F, then there exists $g \in C^\infty(M)$ that $f = g$ on F.*

Theorem 4.17 (Existence of Smooth Exhaustion Function)**.** *Every smooth manifold with or without boundary admits a smooth positive exhaustion function.*

Proof. Let M be a smooth manifold with or without boundary, let $\{V_j\}_{j=1}^\infty$ be any countable open

cover of M by precompact open subsets, and let $\{\psi_j\}$ be a smooth partition of unity subordinate to this cover. Define $f \in C^\infty(M)$ by

$$f(p) = \sum_{j=1}^{\infty} j\psi_j(p)$$

Then f is smooth because only finitely many terms are nonzero in a neighborhood of any point, and positive because $f(p) \geq \sum_j \psi_j(p) = 1$.

To see that f is an exhaustion function, let $c \in \mathbb{R}$ be arbitrary, and choose a positive integer $N > c$. If $p \notin \bigcup_{j=1}^N \bar{V}_j$, then $\psi_j(p) = 0$ for $1 \leq j \leq N$, so

$$f(p) = \sum_{j=N+1}^{\infty} j\psi_j(p) \geq \sum_{j=N+1}^{\infty} N\psi_j(p) = N\sum_{j=1}^{\infty} \psi_j(p) = N > c.$$

Equivalently, if $f(p) \leq c$, then $p \in \bigcup_{j=1}^N \bar{V}_j$. Thus $f^{-1}((-\infty, c])$ is a closed subset of the compact set $\bigcup_{j=1}^N \bar{V}_j$ and is therefore compact. □

Theorem 4.18 (Level Sets of Smooth Functions)**.** *Let M be a smooth manifold. If K is any closed subset of M, there is a smooth nonnegative function $f : M \to \mathbb{R}$ such that $f^{-1}(0) = K$.*

Proof. We begin with the special case in which $M = \mathbb{R}^n$ and $K \subseteq \mathbb{R}^n$ is a closed subset. For each $x \in M \backslash K$, there is a positive number $r \leq 1$ such that $B_r(x) \subseteq M \backslash K$. By Proposition A.16, $M \backslash K$ is the union of countably many such balls $\{B_{r_i}(x_i)\}_{i=1}^\infty$.

Let $h : \mathbb{R}^n \to \mathbb{R}$ be a smooth bump function that is equal to 1 on $\bar{B}_{1/2}(0)$ and supported in $B_1(0)$. For each positive integer i, let $C_i \geq 1$ be a constant that bounds the absolute values of h and all of its partial derivatives up through order i. Define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \sum_{i=1}^{\infty} \frac{(r_i)^i}{2^i C_i} h\left(\frac{x - x_i}{r_i}\right)$$

The terms of the series are bounded in absolute value by those of the convergent series $\sum_i 1/2^i$, so the entire series converges uniformly to a continuous function by the Weierstrass M-test. Because the i th term is positive exactly when $x \in B_{r_i}(x_i)$, it follows that f is zero in K and positive elsewhere.

It remains only to show that f is smooth. We have already shown that it is continuous, so suppose $k \geq 1$ and assume by induction that all partial derivatives of f of order less than k exist and are continuous. By the chain rule and induction, every k th partial derivative of the i th term in the series can be written in the form

$$\frac{(r_i)^{i-k}}{2^i C_i} D_k h\left(\frac{x - x_i}{r_i}\right),$$

where $D_k h$ is some k th partial derivative of h. By our choices of r_i and C_i, as soon as $i \geq k$, each of these terms is bounded in absolute value by $1/2^i$, so the differentiated series also converges uni-

formly to a continuous function. It then follows from Theorem C. 31 that the k th partial derivatives of f exist and are continuous. This completes the induction, and shows that f is smooth.

Now let M be an arbitrary smooth manifold, and $K \subseteq M$ be any closed subset. Let $\{B_\alpha\}$ be an open cover of M by smooth coordinate balls, and let $\{\psi_\alpha\}$ be a subordinate partition of unity. Since each B_α is diffeomorphic to $\mathbb{R}^n$, the preceding argument shows that for each α there is a smooth nonnegative function $f_\alpha : B_\alpha \to \mathbb{R}$ such that $f_\alpha^{-1}(0) = B_\alpha \cap K$. The function $f = \sum_\alpha \psi_\alpha f_\alpha$ does the trick. □

§5 In paracompact Hausdorff space

Lemma 5.1 (Shrinking lemma)**.** *Let X be a paracompact Hausdorff space; let $\{U_\alpha\}_{\alpha \in J}$ be open cover of X. Then there exists a locally finite open cover $\{V_\alpha\}_{\alpha \in J}$ of X such that $\overline{V}_\alpha \subset U_\alpha$ for each α.*

Proof. Let $\mathcal{A}$ be the collection of all open sets A such that $\overline{A}$ is contained in some element of the collection $\{U_\alpha\}$. Regularity of X implies that $\mathcal{A}$ covers X. Since X is paracompact, we can find a locally finite collection $\mathcal{B} = \{B_\beta\}_{\beta \in K}$ of open sets covering X that refines $\mathcal{A}$

Let us index $\mathcal{B}$ bijectively with some index set K, then the general element of $\mathscr{B}$ can be denoted B_β, for $\beta \in K$, and $\{B_\beta\}_{\beta \in K}$ is a locally finite indexed famuly. Since $\mathscr{B}$ refines $\mathscr{A}$, we can define a function $f : K \to J$ by choosing, for each β in K, an element $f(\beta) \in J$ such that

$$\overline{B}_\beta \subset U_{f(\beta)}$$

Then for each $\alpha \in J$, we define V_α to be the union of many B_β

$$V_\alpha = \bigcup_{f(\beta)=\alpha} B_\beta$$

Because the collection $\mathcal{B}_\alpha$ is locally finite, $\overline{V}_\alpha = \bigcup \overline{B}_\beta$, so that $\overline{V}_\alpha \subset U_\alpha$.

Finally, we check local finiteness. Given $x \in X$, choose a neighborhood W of x such that W intersects B_β for only finitely many values of β, say $\beta = \beta_1, \ldots, \beta_K$. Then W can intersect V_α only if α is one of the indices $f(\beta_1), \ldots, f(\beta_K)$.

Theorem 5.2. *Let X be a paracompact Hausdorff space; let $\{U_\alpha\}_{\alpha \in J}$ be an indexed open covering of X. Then there exists a partition of unity on X subordinate to $\{U_\alpha\}$.*

Proof. We begin by applying the shrinking lemma twice, to find locally finite indexed famles of open sets $\{W_\alpha\}$ and $\{V_\alpha\}$ covering X, such that

$$W_\alpha \subset \overline{W}_\alpha \subset V_\alpha \subset \overline{V}_\alpha \subset U_\alpha$$

for each α Since X is normal, we may choose, for each α, a continuous function $\varphi_\alpha : X \to [0,1]$ such that $\varphi_\alpha\left(\overline{W}_\alpha\right) = \{1\}$ and $\varphi_\alpha\left(X - V_\alpha\right) = \{0\}$. Since φ_α is nonzero only at points of V_α, we

have

$$\operatorname{supp} \varphi_\alpha \subset \overline{V}_\alpha \subset U_\alpha$$

Furthermore, the indexed family $\{\overline{V}_\alpha\}$ *is locally finite (since an open set intersects* $\overline{V}_\alpha$ *only if it intersects* V_α *); hence the indexed family* $\operatorname{supp} \varphi_\alpha$ *is also locally finite. Note that because* $\{W_\alpha\}$ *covers* X*, for any given* x *at least one of the functions* ψ_α *is positive at* x*.*

We can now make sense of the formally infinite sum

$$\varphi(x) = \sum_\alpha \varphi_\alpha(x)$$

and define

$$\rho_\alpha(x) = \frac{\varphi_\alpha(x)}{\varphi(x)}$$

to obtain our desired partition of unity.

Chapter II

Tangent Vectors

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§1 The Tangent Space at a Point

Definition 1.1. *Let M be a C^∞ manifold, p be a point in M.*

1. *The **germ** of a C^∞ function at p in M is defined*

$$C_p^\infty(M) := \varinjlim_{U \ni p} C^\infty(U)$$

2. *A **tangent vector** (or say **derivation**) at p is a derivation of $\mathbb{R}$-algebra $C_p^\infty(M)$ (a linear map $D : C_p^\infty(M) \to \mathbb{R}$ such that $D(fg) = (Df)g(p) + f(p)Dg$)*

3. *The tangent vectors at p form a vector space*

$$T_p(M) := \mathrm{Der}_{\mathbb{R}}\left(C_p^\infty(M), \mathbb{R}\right)$$

*called the **tangent space** of M at p. We also write T_pM instead of $T_p(M)$.*

§2 The Differential of a Smooth Map

Definition 2.1. *Let $F : N \to M$ be smooth map between manifolds and a point $p \in N$. We define a map induced by F on the tangent spaces*

$$dF_p : T_pN \to T_{F(p)}M$$

as follows. Given $v \in T_pN$, we let $dF_p(v)$ be the derivation at $F(p)$ that acts on $C^\infty_{F(p)}(M)$ by the rule

$$\langle dF_p(v), \varphi \rangle = \langle v, \varphi \circ F \rangle \quad \text{for all } \varphi \in C^\infty_{F(p)}(M)$$

Proposition 2.2. *Let M, N, and P be smooth manifolds with or without boundary, let $F : M \to N$ and $G : N \to P$ be smooth maps, and let $p \in M$.*

1. *$dF_p : T_pM \to T_{F(p)}N$ is linear.*
2. *The chain rule. $d(G \circ F)_p = dG_{F(p)} \circ dF_p : T_pM \to T_{G\circ F(p)}P$.*
3. *$d\left(\mathrm{Id}_M\right)_p = \mathrm{Id}_{T_pM} : T_pM \to T_pM$.*
4. *If F is a diffeomorphism, then $dF_p : T_pM \to T_{F(p)}N$ is an isomorphism of vector spaces, and $(dF_p)^{-1} = d\left(F^{-1}\right)_{F(p)}$.*

§3 Computations in Coordinates

Proposition 3.1. *Let $(U, \phi) = (U, x^1, \ldots, x^n)$ be a chart about a point p in a manifold M. Then*

1. $$d\phi\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \left.\frac{\partial}{\partial r^i}\right|_{\phi(p)}$$
2. *If $(U, \phi) = (U, x^1, \ldots, x^n)$ is a chart containing p, then the tangent space T_pM has basis*
$$\left.\frac{\partial}{\partial x^1}\right|_p, \ldots, \left.\frac{\partial}{\partial x^n}\right|_p$$
3. *(Transition matrix for coordinate vectors). Suppose $(U, x^1, \ldots, x^n)$ and $(V, y^1, \ldots, y^n)$ are two coordinate charts on a manifold M. Then*
$$\frac{\partial}{\partial x^j} = \sum_i \frac{\partial y^i}{\partial x^j}\frac{\partial}{\partial y^i}$$
on $U \cap V$.

§3.1 The Differential in Coordinates

§3.2 Change of Coordinates

§4 The Tangent Bundle

Definition 4.1. *Given a smooth manifold M with or without boundary, we define the* ***tangent bundle*** *of M, denoted by TM, to be the disjoint union of the tangent spaces at all points of M :*

$$TM = \coprod_{p \in M} T_p M$$

We usually write an element of this disjoint union as an ordered pair (p, v) of v_p, with $p \in M$ and $v \in T_p M$. The tangent bundle comes equipped with a ***natural projection map*** *$\pi : TM \to M$, which sends each vector in $T_p M$ to the point p at which it is tangent: $\pi(p, v) = p$.*

Proposition 4.2. *For any smooth n-manifold M, the tangent bundle TM has a natural topology and smooth structure that make it into a $2n$-dimensional smooth manifold. With respect to this structure, the projection $\pi : TM \to M$, $(x^i, v^i) \mapsto (x^i)$ is smooth.*

Proof. We begin by defining the maps that will become our smooth charts. Given any smooth chart $(U, \varphi, x^1, \ldots, x^n)$ for M and define a injective map $\tilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2n}$ by

$$\tilde{\varphi}\left(v^i \frac{\partial}{\partial x^i}\bigg|_p \right) = \left(x^1(p), \ldots, x^n(p), v^1, \ldots, v^n \right)$$

Now suppose we are given two smooth charts (U, φ) and (V, ψ), and let $(\pi^{-1}(U), \tilde{\varphi}), (\pi^{-1}(V), \tilde{\psi})$ be the corresponding charts on TM. The sets

$$\begin{aligned} \tilde{\varphi}\left(\pi^{-1}(U) \cap \pi^{-1}(V)\right) &= \varphi(U \cap V) \times \mathbb{R}^n \quad \text{and} \\ \tilde{\psi}\left(\pi^{-1}(U) \cap \pi^{-1}(V)\right) &= \psi(U \cap V) \times \mathbb{R}^n \end{aligned}$$

are open in $\mathbb{R}^{2n}$, and the transition map $\tilde{\psi} \circ \tilde{\varphi}^{-1} : \varphi(U \cap V) \times \mathbb{R}^n \to \psi(U \cap V) \times \mathbb{R}^n$ can be written explicitly as

$$\begin{aligned} &\tilde{\psi} \circ \tilde{\varphi}^{-1}\left(x^1, \ldots, x^n, v^1, \ldots, v^n\right) \\ &= \left(\tilde{x}^1(x), \ldots, \tilde{x}^n(x), \frac{\partial \tilde{x}^1}{\partial x^j}(x) v^j, \ldots, \frac{\partial \tilde{x}^n}{\partial x^j}(x) v^j \right). \end{aligned}$$

This is clearly smooth.

Choosing a countable cover $\{U_i\}$ of M by smooth coordinate domains, we obtain a countable cover of TM by coordinate domains $\{\pi^{-1}(U_i)\}$ satisfying conditions (i)-(iv) of the smooth manifold chart lemma (Lemma 1.35). To check the Hausdorff condition (v), just note that any two points in the same fiber of π lie in one chart, while if (p, v) and (q, w) lie in different fibers, there exist

disjoint smooth coordinate domains U, V for M such that $p \in U$ and $q \in V$, and then $\pi^{-1}(U)$ and $\pi^{-1}(V)$ are disjoint coordinate neighborhoods containing (p, v) and (q, w), respectively.

To see that π is smooth, note that with respect to charts (U, φ) for M and $(\pi^{-1}(U), \tilde{\varphi})$ for TM, its coordinate representation is $\pi(x, v) = x$.

The coordinates (x^i, v^i) are called **natural coordinates** on TM. □

§5 Curves in a Manifold and Velocity Vectors

Definition 5.1. *If M is a manifold with or without boundary, we define a **curve** in M to be a continuous map $\gamma : J \to M$; where $J \subset \mathbb{R}$ is an interval.*

Definition 5.2. *Now let M be a smooth manifold with or without boundary. Our definition of tangent spaces leads to a natural interpretation of velocity vectors: given a smooth curve $\gamma : J \to M$ and $t_0 \in J$, we define the **velocity** of γ at t_0, denoted by $\gamma'(t_0)$, to be the vector*

$$\gamma'(t_0) = d\gamma\left(\left.\frac{d}{dt}\right|_{t_0}\right) \in T_{\gamma(t_0)}M$$

where $d/\,dt|_{t_0}$ is the standard coordinate basis vector in $T_{t_0}\mathbb{R}$.

This tangent vector acts on functions by

$$\gamma'(t_0)\, f = d\gamma\left(\left.\frac{d}{dt}\right|_{t_0}\right) f = \left.\frac{d}{dt}\right|_{t_0}(f \circ \gamma) = (f \circ \gamma)'(t_0)$$

In other words, $\gamma'(t_0)$ is the derivation at $\gamma(t_0)$ obtained by taking the derivative of a function along γ. (If t_0 is an endpoint of J, this still holds, provided that we interpret the derivative with respect to t as a one-sided derivative, or equivalently as the derivative of any smooth extension of $f \circ \gamma$ to an open subset of $\mathbb{R}$.)

Now let (U, φ, x^i) be a smooth chart. If $\gamma(t_0) \in U$, we can write the coordinate representation of γ as $\gamma(t) = (\gamma^1(t), \ldots, \gamma^n(t))$ for t sufficiently close to t_0, and then the coordinate formula for the differential yields

$$\gamma'(t_0) = \frac{d\gamma^i}{dt}(t_0)\left.\frac{\partial}{\partial x^i}\right|_{\gamma(t_0)}$$

Proposition 5.3 (The Velocity of a Composite Curve)**.** *Let $F : M \to N$ be a smooth map, and let $\gamma : J \to M$ be a smooth curve. For any $t_0 \in J$, the velocity at $t = t_0$ of the composite curve $F \circ \gamma : J \to N$ is given by*

$$(F \circ \gamma)'(t_0) = dF(\gamma'(t_0))$$

Proof. Just go back to the definition of the velocity of a curve:

$$(F \circ \gamma)'(t_0) = d(F \circ \gamma)\left(\left.\frac{d}{dt}\right|_{t_0}\right) = dF \circ d\gamma\left(\left.\frac{d}{dt}\right|_{t_0}\right) = dF(\gamma'(t_0))$$

□

Corollary 5.4 (Computing the Differential Using a Velocity Vector)**.** *Suppose* $F : M \to N$ *is a smooth map,* $p \in M$*, and* $v \in T_pM$*. Then*

$$dF_p(v) = (F \circ \gamma)'(0)$$

for any smooth curve $\gamma : J \to M$ *such that* $0 \in J, \gamma(0) = p$*, and* $\gamma'(0) = v$*.*

Chapter III

Rank

Definition 0.1. *Let $F : N \to M$ be a smooth map between smooth manifolds with or without boundary, and let $p \in N$.*

1. *We define the rank of $\boldsymbol{F}$ at $\mathbf{p}$ to be the rank of the linear map $dF_p : T_pN \to T_{F(p)}M$.*
2. *If the rank of dF_p is equal to this upper bound, we say that $\boldsymbol{F}$ has full rank at $\mathbf{p}$, and if F has full rank everywhere, we say F has full rank.*
3. *f is said **submersion** at p if $\mathrm{d}f_p$ is surjective.*

   *f is said **inmersion** if $\mathrm{d}f_p$ is injective*

§1 Constant Rank Theorem

Theorem 1.1 (Constant rank theorem)**.** *Let N and M be manifolds of dimensions n and m respectively. Suppose $f : N \to M$ has constant rank k in a neighborhood of a point p in N. Then there are charts (U, ϕ) centered at p in N and (V, ψ) centered at $f(p)$ in M such that for $(r^1, \ldots, r^n)$ in $\phi(U)$,*

$$\left(\psi \circ f \circ \phi^{-1}\right)\left(r^1, \ldots, r^n\right) = \left(r^1, \ldots, r^k, 0, \ldots, 0\right).$$

Theorem 1.2 (Constant-rank level set theorem)**.** *Let $f : N \to M$ be a C^∞ map of manifolds and $c \in M$. If f has constant rank k in a neighborhood of the level set $f^{-1}(c)$ in N, then $f^{-1}(c)$ is a regular submanifold of N of codimension k.*

Proof. Let p be an arbitrary point in $f^{-1}(c)$. By the constant rank theorem there are a coordinate chart $(U, \phi) = (U, x^1, \ldots, x^n)$ centered at $p \in N$ and a coordinate chart $(V, \psi) = (V, y^1, \ldots, y^m)$ centered at $f(p) = c \in M$ such that

$$\left(\psi \circ f \circ \phi^{-1}\right)\left(r^1, \ldots, r^n\right) = \left(r^1, \ldots, r^k, 0, \ldots, 0\right) \in \mathbb{R}^m.$$

This shows that the level set $(\psi \circ f \circ \phi^{-1})^{-1}(0)$ is defined by the vanishing of the coordinates $r^1, \ldots, r^k$. The image of the level set $f^{-1}(c)$ under ϕ is the level set $(\psi \circ f \circ \phi^{-1})^{-1}(0)$ (Figure 11.1), since

$$\phi\left(f^{-1}(c)\right) = \phi\left(f^{-1}\left(\psi^{-1}(0)\right) = \left(\psi \circ f \circ \phi^{-1}\right)^{-1}(0).$$

Thus, the level set $f^{-1}(c)$ in U is defined by the vanishing of the coordinate functions $x^1, \ldots, x^k$, where $x^i = r^i \circ \phi$. This proves that $f^{-1}(c)$ is a regular submanifold of N of codimension k. $\square$

Chapter IV

Sard's Theorem

§1 Sets of Measure Zero

Definition 1.1. *If M is a smooth m-manifold with or without boundary, we say that a subset $A \subseteq M$* ***has measure zero in*** *M if for every smooth chart (U, φ) for M, the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n-dimensional measure zero.*

Lemma 1.2. *Let M be a smooth n-manifold with or without boundary and $A \subseteq M$. Suppose that for some collection $\{(U_\alpha, \varphi_\alpha)\}$ of smooth charts whose domains cover $A, \varphi_\alpha(A \cap U_\alpha)$ has measure zero in $\mathbb{R}^n$ for each α. Then A has measure zero in M.*

Proof: Let (V, ψ) be an arbitrary smooth chart. We need to show that $\psi(A \cap V)$ has measure zero. Some countable collection of the U_α 's covers $A \cap V$. For each such U_α, we have

$$\psi(A \cap V \cap U_\alpha) = \left(\psi \circ \varphi_\alpha^{-1}\right) \circ \varphi_\alpha(A \cap V \cap U_\alpha)$$

Now, $\varphi_\alpha(A \cap V \cap U_\alpha)$ is a subset of $\varphi_\alpha(A \cap U_\alpha)$, which has measure zero in $\mathbb{R}^n$ by hypothesis. By Proposition 6.5 applied to $\psi \circ \varphi_\alpha^{-1}$, therefore, $\psi(A \cap V \cap U_\alpha)$ has measure zero. Since $\psi(A \cap V)$ is the union of countably many such sets, it too has measure zero.

Theorem 1.3. *Suppose M and N are differential m-manifolds with or without boundary, $F : M \to N$ is a C^1 map, and $A \subseteq M$ is a subset of measure zero. Then $F(A)$ has measure zero in N.*

Corollary 1.4. *Suppose M and N are differential manifolds with or without boundary, $\dim M \leq \dim N$, and $F \in C^1(M, N)$. If $A \subset M$ is a subset of measure zero, then $F(A)$ has measure zero in N.*

§2 Sard's Theorem

Definition 2.1. *If $f : M \to N$ is a smooth map,*

(1) A point $p \in M$ is said to be a ***regular point of*** *f if $df_p : T_pM \to T_{f(p)}N$ is surjective ($\operatorname{rank}_p f = \dim N$). A point $c \in N$ is said to be a* ***regular value*** *of f if every point of the level set $f^{-1}(c)$ is a regular point.*

(2) A point $q \in M$ is said to be a ***critical point of*** *f if $df_p : T_pM \to T_{f(p)}N$ is not surjective* ($\operatorname{rank}_p f < \dim N$). *A point $d \in N$ is said to be a* ***critical value*** *of f if there exists a critical point in level set $f^{-1}(d)$.*

Theorem 2.2 (Sard's Theorem)**.** *Suppose M and N are smooth manifolds with or without boundary and $f : M \to N$ is a smooth map. Then the set of critical values of f has measure zero in N.*

Corollary 2.3. *Suppose M and N are smooth manifolds with or without boundary, and $F : M \to N$ is a smooth map. If* $\dim M < \dim N$*, then $F(M)$ has measure zero in N.*

Corollary 2.4. *Suppose M is a smooth manifold with or without boundary, and $S \subseteq M$ is an immersed submanifold with or without boundary. If* $\dim S < \dim M$*, then S has measure zero in M.*

§3 The Whitney Embedding Theorem

Theorem 3.1 (Whitney Embedding Theorem)**.** *Every smooth n-manifold with or without boundary admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.*

Corollary 3.2. *Every smooth n-dimensional manifold with or without boundary is diffeomorphic to a properly embedded submanifold (with or without boundary) of $\mathbb{R}^{2n+1}$.*

Corollary 3.3. *Suppose M is a compact smooth n-manifold with or without boundary. If $N \geq 2n+1$, then every smooth map from M to $\mathbb{R}^N$ can be uniformly approximated by embeddings.*

Theorem 3.4 (Whitney Immersion Theorem)**.** *Every smooth n-manifold with or without boundary admits a smooth immersion into $\mathbb{R}^{2n}$.*

Theorem 3.5 (Strong Whitney Immersion Theorem)**.** *If $n > 1$, every smooth n-manifold admits a smooth immersion into $\mathbb{R}^{2n-1}$.*

Chapter V

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§1

Definition 1.1. *Let X, B, and F be Hausdorff spaces and $p : X \to B$ a map. Then p is called a* ***bundle projection with fiber*** *F, if each point of B has a neighborhood U such that there is a homeomorphism*

$$\phi : U \times F \to p^{-1}(U), \quad \textit{that } p(\phi\langle b, y\rangle) = b$$

for all $b \in U$ and $y \in F$. Such a map ϕ is called a ***trivialization of the bundle over*** *U.*

Definition 1.2. *Let K be a topological group acting effectively on the Hausdorff space F as a group of homeomorphisms. Let X and B be Hausdorff spaces. By a fiber bundle over the base space B with total space X, fiber F, and structure group K, we mean a bundle projection $p : X \to B$ together with a collection $\mathcal{A}$ of trivializations $\phi : U \times F \to p^{-1}(U)$, of p over U, called charts over U, such that:*

(i) each point of B has a neighborhood over which there is a chart in Φ;

(ii) if $\phi : U \times F \to p^{-1}(U)$ is in $\mathcal{A}$ and $V \subset U$ then the restriction of ϕ to $V \times F$ is in $\mathcal{A}$

(iii) if $\phi, \psi \in \mathcal{A}$ are charts over U then there is a map $\theta : U \to K$ such that $\psi\langle u, y\rangle = \phi\langle u, \theta(u)(y)\rangle$; and

(iv) the set $\mathcal{A}$ is maximal among collections satisfying (a), (b), and (c).

The bundle is called smooth if all these spaces are manifolds and all maps involved are smooth.

Definition 1.3. *A* ***vector bundle*** *is a fiber bundle in which the fiber is a euclidean space and the structure group is the general linear group of this euclidean space or some subgroup of that group.*

A vector bundle is usually denoted by a Greek letter such as ξ and its total space by $E(\xi)$ and base space by $B(\xi)$. Its fiber projection is denoted by π_ξ or just by π. The following definition, given only for vector bundles, has a fairly obvious generalization to general fiber bundles, but we need it only for vector bundles.

Definition 1.4. *If ξ and η are vector bundles then a* ***bundle map*** *$\xi \to \eta$ is a map $g : E(\xi) \to E(\eta)$ carrying each fiber of ξ onto some fiber of η isomorphically. A bundle map g is a* ***bundle isomorphism*** *or a bundle equivalence if it is a homeomorphism. (In particular, the fibers have the same dimension and there is an induced map $B(\xi) \to B(\eta)$.)*

§2

Definition 2.1. *Let M, X, Y be smooth manifolds and let $f : X \to M$ and $g : Y \to M$ be smooth maps with g an embedding (Y is a submanifold of M). Then f is said to be transverse :o g (denoted by $f \pitchfork g$) if, whenever $f(x) = g(y)$, the images of the differentials $f_* : T_x(X) \to T_{f(x)}(M)$ and $g_* : T_y(Y) \to T_{g(y)}(M) = T_{f(x)}(M)$ span $T_{f(x)}(M)$.*

Definition 2.2. *Let $f : M \to N$ be a smooth map, and $X \subset N$ be a smooth submanifold. We say f* ***intersect*** *X* ***transversally****, and denote by $f \pitchfork g$, if*

$$\mathrm{Im}\,(df_p) + T_{f(p)}X = T_{f(p)}N, \quad \forall p \in f^{-1}(X)$$

Definition 2.3. *We say two smooth submanifolds X_1 and X_2 in M* ***intersect transversally*** *if for any $p \in X_1 \cap X_2$,*

$$T_pX_1 + T_pX_2 = T_pM$$

In this case we write $X_1 \pitchfork X_2$.